Reference
https://mathworld.wolfram.com/Point-PlaneDistance.html

1 vectors and planes review

1A)
Assume x1 & x2 represent the numbered 3-dimensional vectors.

Expression for calculating the angle between the two vectors is $\theta = \frac{x1 \cdot x2}{||x1||||x2||}$
The two vectors are orthogonal to one another when their dot product is 0.
$x1 \cdot x2 = 0$
1B)
$2\vec{\theta}$ as an example. There are infinite alternative possible vectors, because regardless of magnitude (except 0) the vector will always describe hyperplane p because it represents the normal vector. That means there are $a\vec{\theta}$ vectors for any scalar $a \geq 0$ so that the direction is not changed.
1C)

(i) Let p_1 be the plane consisting of the set of points for which $x_1 - 2x_2 = 0$. What is the signed distance of point $\vec{a} = [-1, 1]^T$ from p_1?

normal vector: $\vec{\theta} = [1, -2]^T$
vector from plane to point: $\vec{w} = [x_1 + 1, x_2 - 1]^T$
and the signed distance is the projection of $\vec{w}$ onto $\vec{\theta}$: $\frac{|\vec{\theta} \cdot \vec{w}|}{||\vec{\theta}||}$

$$|\vec{\theta} \cdot \vec{w}| = |1(x_1 + 1) - 2(x_2 - 1)| = |x_1 + 1 - 2x_2 + 2| = 3$$

$$||\vec{\theta}|| = \sqrt{1^2 + (-2)^2} = \sqrt{5}$$

$d = \frac{3}{\sqrt{5}} = \frac{3\sqrt{5}}{5}$

d is the signed distance of point $\vec{a}$ to p_1
getting the

(ii) Let p_2 be the plane consisting of the set of points for which $x_1 - 2x_2 + 5 = 0$. What is the signed distance of point $\vec{a} = [-1, 1]^T$ from p_2?

normal vector: $\vec{\theta} = [1, -2]^T$

$d = \frac{-1-2+5}{\sqrt{(1)^2+(-2)^2+(5)^2}} = \frac{2}{\sqrt{30}} = \frac{\sqrt{30}}{15}$

1D)
No, if a point is projected onto a plane, the original point cannot be recovered by projecting back, because information is lost during projection.

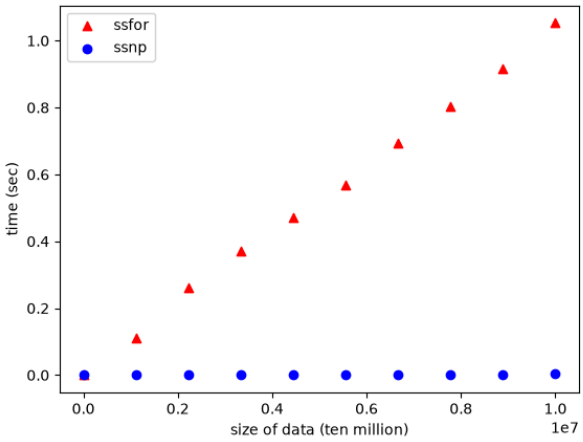
Assume plane with normal vector $\vec{\theta} = [1, 1]$ with offset b = -1. Let points a_1& a_2 be projected onto this plane (may use a1 and a2 interchangeably.)

projection equation is $proj_a d = \frac{a \cdot d}{||d||^2} d$
in which a are the points that are projected onto d the plane.

2 linear algebra review

2A)
(i) $\vec{v} \cdot \vec{v}$
(ii) $||\vec{v} \cdot \vec{v}||cos(\theta)$
(iii) $\vec{v} \times \vec{v}^T$
2B) $\sqrt{||\vec{\theta}X - \vec{y}||}$
2C) true.
If AB & BA are both defined then the matrices must be square matrices to be two-way compatible for multiplication. Let A be a matrix of [a1, a2, …, an] B similarly with [b1,b2,..bn] then AB = [a1b1,a2b2, …, anbn] and (AB)^T = [a1b1,a2b2, …, anbn]^T. Individually A^T will also result in [a1, a2, …, an]^T and same for B which multiplied together will also result in (AB)^T.
2D) No it is not possible.
Because rank(C) ≤ min (rank(A), rank(B))
Therefore, because B is of pxn it's highest rank can only be p. Since
p<n, the highest rank C can achieve is the min of either rank A (possible n) or rank B (highest possible p) which means the highest rank C can be is p, so rank(C) can't be n.

3 vectorization comparison



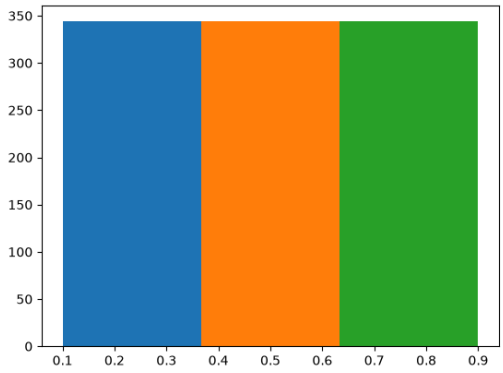

```
"""
code for the graph (3F) below
"""
#run time_ss of values from 10 to 10M
#rn test will b small to easily run at first
# 10 000 000
samples = np.linspace(10,10000000,10, dtype=int)
#np.arange(10, 10000000).tolist()
#np.linspace(10,10000000,3, dtype=int)
tmp_dict = time_ss(samples)
df = timess_to_df(tmp_dict)

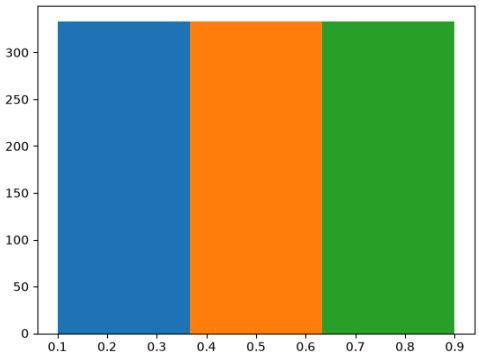

#have x be the size of list
#have y be the time for
x = df['n']
y1 = df['ssfor']
y2 = df['ssnp']
plt.scatter(x,y1,c = "red",marker="^", label= "ssfor")
plt.scatter(x,y2,c = "blue", marker= "o",label= "ssnp")
plt.xlabel('size of data (ten million)')
plt.ylabel('time (sec)')
plt.legend()
plt.show()
```

5 visualization exploration

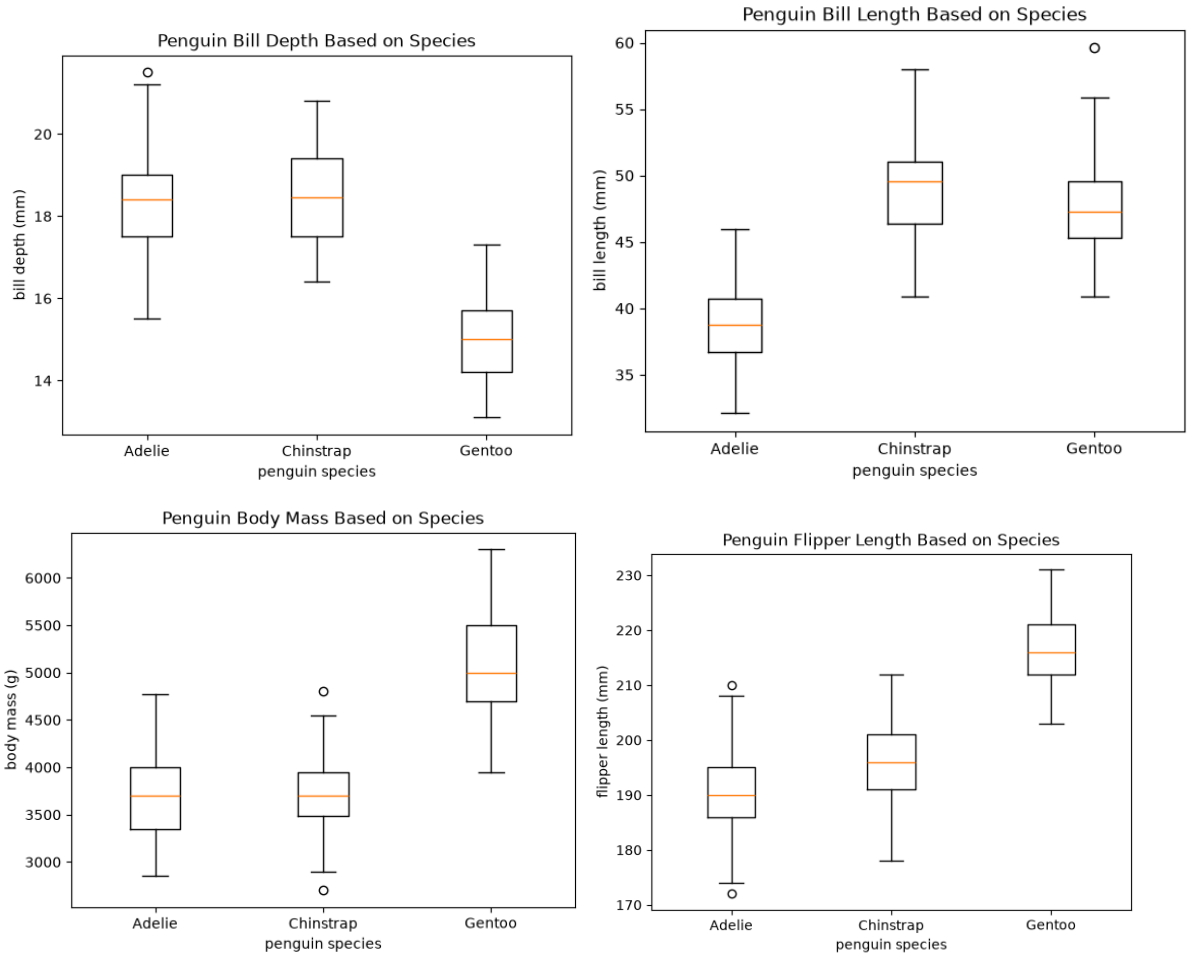
5A)

No, dropping the NaN values does not change the distribution of the species. It can be seen in these histograms. The one on the left containing the NaN values and the right without. They aren't labeled but the different colors refer to the different species.

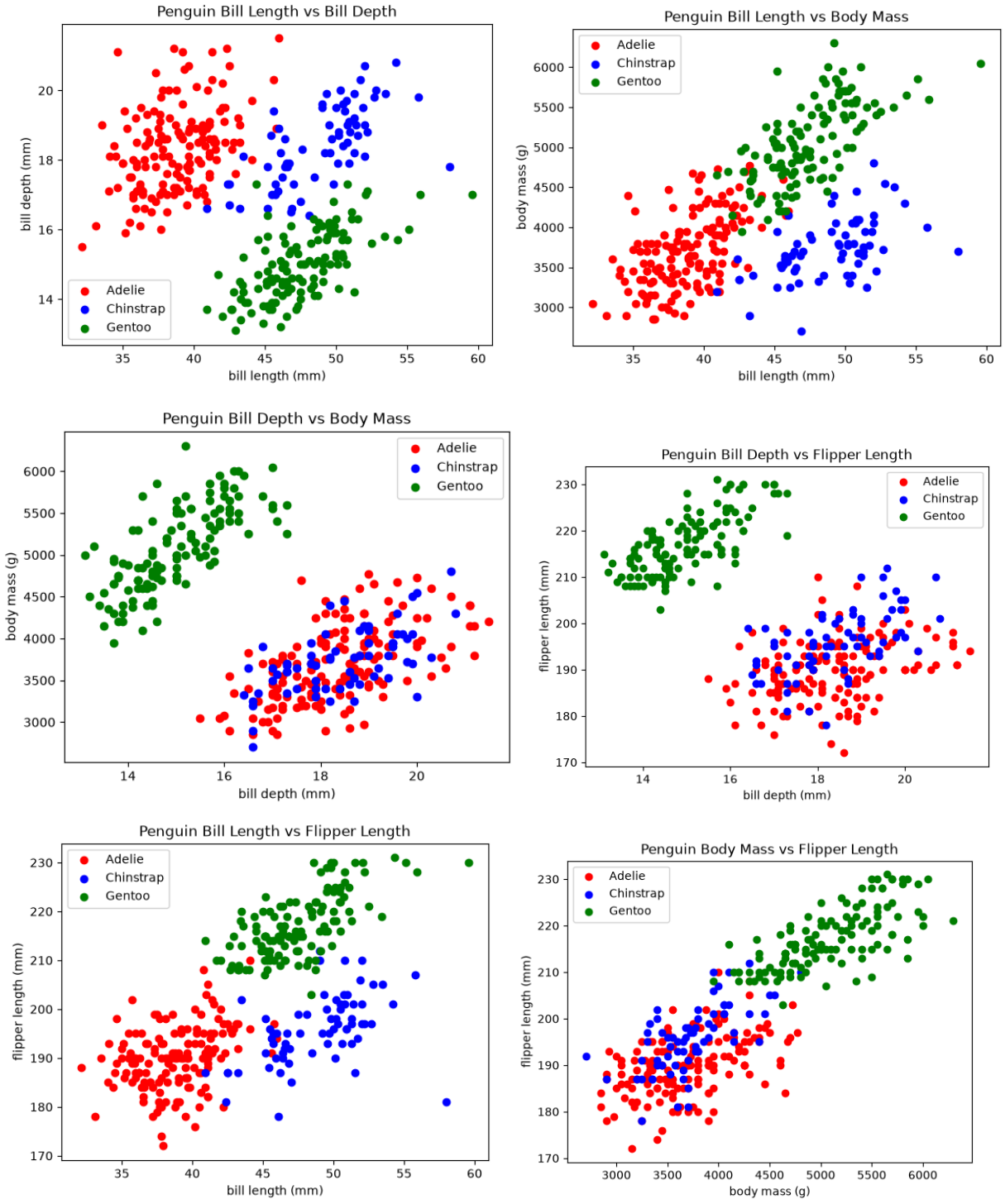




5B)


Penguin Bill Depth Based on Species
bill depth (mm)
20
18
16
14
Adelie
Chinstrap
Gentoo
penguin species
Penguin Bill Length Based on Species
bill length (mm)
60
55
50
45
40
35
Adelie
Chinstrap
Gentoo
penguin species
Penguin Body Mass Based on Species
body mass (g)
6000
5500
5000
4500
4000
3500
3000
Adelie
Chinstrap
Gentoo
penguin species
Penguin Flipper Length Based on Species
flipper length (mm)
230
220
210
200
190
180
170
Adelie
Chinstrap
Gentoo
penguin species

5C)



5D)

Adelie and Chinstrap Penguins tend to have more bill depth compared to that of Gentoos. The Adelie's bill length is shorter compared to that of the Chinstraps and Gentoos. This can be seen with the top left and top right box plots respectively. In which the means of the bill depth for Adelies and Chinstraps are around 18-19 mm range and the gentoos are around the 15mm range.

Likewise the mean for the Adelie penguin bill length is around 38 and the mean for the other penguins are in the 47-50 mm range.
Gentoos also have more body mass and longer flipper length compared to that of Adelie and Chinstrap penguins. As seen in the data from the last two boxplots. In which the gentoos mean always exceeds that of the upper limit for the other two species.
Distinctually, Gentoos are midranged in bill length and have largest body mass, whereas Adelie penguins have the lowest for both, and Gentoo penguins have equally low body mass but generally a higher bill length. According to the top right Bill Length vs Body Mass scatter plot data.
Adelies and Chinsraps tend to be more common in regards to body mass, flipper length, and bill depth than that of the Gentoo which can be seen in the 3rd, 4th and 6th scatter plot data in which the data points are more closely clustered and the gentoo scatters are off to another cluster region.